

# Forside

## Eksamensinformation

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## Besvarelsen afleveres af

Jakob Hellesøe Mathiesen  
fdb716@alumni.ku.dk

## Eksamensadministratorer

### Bedømmere

Mikkel Godt Gregersen  
Eksaminator  
mikkel.godt.gregersen@econ.ku.dk

Christian Farø  
Censor  
cf@business.aau.dk  
 +4521467940

## Besvarelsesinformationer

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# Valuation of Novo Nordisk A/S

With Emphasis on the Cost of Capital

**Seminar: Anvendt Corporate Finance**  
Jakob Hellesøe Mathiesen

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# Valuation of Novo Nordisk A/S

**Jakob Hellesøe Mathiesen**

University of Copenhagen

Department of Economics

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## Abstract

This paper seeks to determine the intrinsic value of Novo Nordisk A/S as at December 31, 2023. The valuation concentrates on the cost of capital, with emphasis on the cost of equity, due to the capital structure of Novo Nordisk. The paper explores various theoretical approaches for deriving the equity risk premium, beta-value etc. within the framework of the CAPM. Applying the WACC formula, the cost of capital is estimated to 6.4%, with a cost of equity of 6.43% and a cost of debt of 3.31%. Ultimately the paper utilizes a DCF model estimating the value of equity at DKK 4,031 bn, yielding a share price of DKK 899. With an actual share price of DKK 698 at valuation date, the estimated price exceeds the actual price by 29% indicating an undervalued stock. A sensitivity analysis is performed to evaluate the robustness of the estimates, from which it is concluded, that even small changes in the parameter values has significant impact on the estimated intrinsic value of the company. This emphasizes the importance of the thorough analysis of the cost of capital to achieve the most accurate and precise valuation.

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# 1 Introduction

As Europe's most valuable company, Novo Nordisk A/S has an immense impact on especially the Danish economy. The company is a global leader in diabetes care and has made significant strides in other therapeutic areas, most notably in obesity treatments. In recent years, Novo Nordisk has experienced impressive global growth, largely driven by the significant advancements in obesity drugs. This success has not only boosted its market capitalization [Yahoo Finance, 2024], but also increased the attention and demand among international investors [Reuters, 2023a].

The remarkable growth trajectory of Novo Nordisk raises important questions about investor sentiment and market valuation. A consideration is whether the strong performance and growth have led to an overly optimistic investor view, potentially inflating the company's market value. Alternatively, the company's growth potential is not yet fully recognized and reflected in its current market valuation. These considerations are particularly relevant for Danish private investors, as the majority of their returns last year came from shares in Novo Nordisk. At the end of 2023, Danes owned a total of DKK 190 billion worth of shares in the company [Børsen, 2024].

In light of Novo Nordisk's growing influence as the most valuable firm in Europe and the continued growth potential, this paper seeks to answer the following research question:

*What is the intrinsic value of Novo Nordisk A/S as of December 31, 2023?*

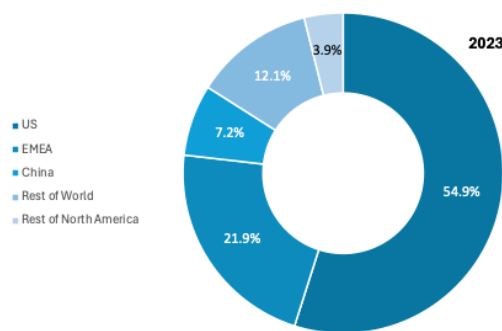
The valuation will employ a Discounted Cash Flow (DCF) model based on forecasted values and will apply the weighted average cost of capital (WACC) as discount rate. A substantial part of this paper will concentrate on evaluating the cost of capital by examining different methodologies and discussing both the consensus and the related disagreements in estimation techniques. The company is essentially equity-financed, hence a significant part of the paper will focus on the cost of equity. The paper recognizes that even small changes in input parameters significantly affect the valuation, hence a sensitivity analysis will be performed to evaluate these effects. No data or information post-valuation date has been used in the valuation.

## 2 Company & Market Presentation

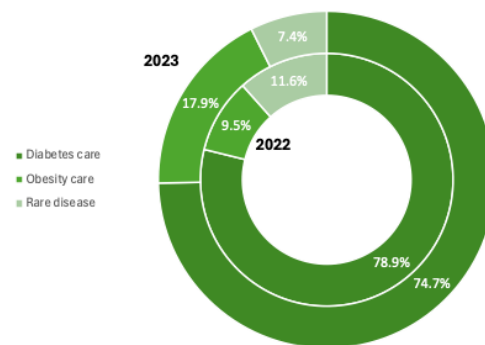
### 2.1 Novo Nordisk A/S

Novo Nordisk A/S, founded in 1923 and headquartered in Denmark, is a leading global healthcare company. Known for its contributions to diabetes care, obesity and rare disease treatments. Novo Nordisk employs over 64,000 people across 80 offices and markets its products in 170 countries [Novo Nordisk]. As shown in Figure 2.1, 54.9% of its 2023 revenue came from the US, the largest market, while EMEA countries accounted for 21.9%.

Novo Nordisk's primary business remains its diabetes care portfolio, with an increasing impact from obesity drugs. In 2023 diabetes care made up 74.7% of the revenue, decreasing by 4.2 pp from 2022. In recent years, Novo Nordisk has introduced innovative solutions in obesity care, which is a rapidly growing segment due to the global increase in obesity rates. Their drug Wegovy was approved by the FDA in 2021 and has gained significant attention for its effectiveness in weight loss.



**Figure 2.1:** Net sales split by countries



**Figure 2.2:** Net sales split by segments

Source: [Annual Report 2023]

### 2.2 The Industry

The diabetes care and obesity management industries are experiencing significant growth and innovation. The global diabetes care market, valued at approximately USD 79 billion in 2023 [Fortune, 2023], is driven by the increasing prevalence of diabetes, advancements in technology, and the rising obesity rates. The market for drugs as insulin is expected to see continued growth due to its essential role in managing diabetes effectively. The key competitors within the market of diabetes care and obesity drugs include Sanofi, Eli Lilly, Novartis AG and others. The obesity management industry is also expanding rapidly, reflecting the global rise in obesity rates, which affects over 890 million adults [WHO, 2023]. This sector has seen substantial investments in innovative treatments, which has been well-received for its effectiveness in weight loss. As the obesity management industry is relatively young, all competitors are actively working on developing innovative treatments to meet the increasing demand.

## 3 Strategic Analysis

### 3.1 PESTEL Analysis

In this section, the PESTEL framework is used to identify key external factors impacting Novo’s performance. While Appendix A.1 provides a comprehensive analysis, Table 3.1 summarizes the most significant elements affecting Novo’s financial drivers.

| Aspect           | Main Points                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Political</b> | Novo Nordisk’s operations are influenced by healthcare regulations, government funding, pricing scrutiny, and international trade policies. U.S. and European regulations impact drug market access and pricing, affecting profitability. Political pressure on drug pricing, such as criticism from Senator Bernie Sanders, poses challenges [MedWatch, 2023]. Future political effects will be mixed, with potential limits on profit margins due to pricing scrutiny but benefits from supportive healthcare policies. Novo Nordisk’s lobbying and partnerships may mitigate negative impacts and secure favorable policies. |
| <b>Social</b>    | Demographic trends like an aging population and rising obesity and diabetes rates expand Novo Nordisk’s potential market. Changing views on obesity as a treatable condition also boost demand. Global projections indicate significant increases in diabetes and obesity cases, with diabetes expected to rise by 19.7% from 537 million in 2021 to 643 million by 2030 [IDF, 2021], and the overweight and obese population projected to increase from 2.6 billion in 2020 to 4.0 billion by 2035 [World Obesity, 2023]. These trends suggest a positive social impact on Novo Nordisk’s future profitability.                |
| <b>Legal</b>     | Intellectual property rights, health regulations, and patent laws shape Novo Nordisk’s strategy. These legal conditions influence clinical trials, innovation protection, and global competition. Drug development is costly and risky. Development takes 10-15 years with only 12% of drugs gaining FDA approval. Patent laws grant temporary monopolies, allowing recovery of R&D costs, but generic competition after patents expire reduces prices and profits. Strong patent protection is crucial for Novo Nordisk’s financial health and ongoing drug development. [PhRMA, 2023]                                         |

**Table 3.1:** Summary of Political, Social, and Legal Factors Affecting Novo Nordisk

### 3.2 Porter’s Five Forces

This section uses Porter’s Five Forces to examine key industry trends affecting Novo’s competitive environment. From the detailed analysis in Appendix A.2, the most critical insights are summarized in Table 3.2.

| Aspect                        | Main Points                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Threat of New Entrants</b> | High entry barriers due to significant capital requirements for production facilities and laboratory equipment, lengthy and costly R&D process, and limited time to recoup investments due to 20-year patents (10-15 years spent on R&D). These factors reduce the risk of new entrants. [FDA, 2015]                                                                                                                                                                                                                            |
| <b>Competitive Rivalry</b>    | Moderate. Limited competition in diabetes and obesity medicine markets due to extensive patent use, creating monopoly-like conditions. Novo Nordisk holds several patents (6 in diabetes care, 1 for obesity drug Wegovy, and 2 in rare diseases) [Annual Report 2023]. Novo and its major competitors (Sanofi and Eli Lilly) control over 90% of the global insulin market [Knox, 2020]. The obesity market is projected to reach USD 77 billion by 2030 [Morgan Stanley, 2023], reducing rivalry by increasing opportunities. |

**Table 3.2:** Summary of Market Forces Affecting Novo Nordisk

## 4 Financial Analysis

### 4.1 Profitability Analysis

This section analyzes Novo Nordisk’s profitability using the extended DuPont model. Utilizing reformulated financial statements from Appendix A.3, the analysis covers a 5-year period to understand the company’s value drivers. The DuPont model decomposes return on common equity into three levels [Sørensen, 2022], which will be examined in the following sections.

#### 4.1.1 Decomposition Level 1

The highest level of the decomposition breaks down the Return on Common Equity (ROCE) into operational and financial activities. The ROCE consist of the Return on Net Operating Assets (RNOA), which is the operational driver, and the financial drivers given by the Financial Leverage (FLEV) and the Net Borrowing Costs (NBC) [Penman, 2013]. The ROCE is defined as follows:

$$\text{ROCE} = \text{RNOA} + [\text{FLEV} \cdot (\text{RNOA} - \text{NBC})] \quad (4.1)$$

As illustrated in Table 4.1, Novo Nordisk’s ROCE has been increasing from 2020-2023 peaking at 88.08%. Compared to the industry average, the relatively high ROCE is primarily driven by RNOA averaging at 81.9% during the past five years. The RNOA has for the entire period significantly outperformed the industry, with the industry averaging at 13.4% in 2023 [Damodaran, 2024c].

|        | 2023   | 2022   | 2021    | 2020    | 2019    |
|--------|--------|--------|---------|---------|---------|
| ROCE   | 88.08% | 72.62% | 69.85%  | 67.93%  | 70.92%  |
| RNOA   | 88.29% | 73.61% | 67.24%  | 79.59%  | 100.76% |
| FLEV   | -2.20% | 6.55%  | 3.14%   | -13.08% | -24.06% |
| NBC    | 78.39% | 88.70% | -15.80% | -9.58%  | -23.29% |
| Spread | 9.90   | -15.08 | 83.04   | 89.18   | 124.05  |

**Table 4.1:** Decomposition Level 1

**Note:** The Spread is denoted in percentage points.

**Source:** Novo Nordisk [Annual Report 2018] - [Annual Report 2023] and own calculations

The increasing FLEV from 2019-2022 indicates growing financial obligations for Novo Nordisk. In 2023, the FLEV was -2.2%, showing net financial assets, similar to 2019-2020. The fluctuating NBC reflects shifts between net financial obligations and assets. With RNOA fairly stable, the spread between RNOA and NBC decreased from 124.05 pp to 9.9 pp over the last five years. In 2023, near-zero FLEV indicates that strong ROCE performance was entirely driven by RNOA.

#### 4.1.2 Decomposition Level 2

The second level decomposition will focus on the drivers of RNOA, which is the Profit Margin (PM) and the Asset Turnover Rate (ATO), and is computed as

$$\text{RNOA} = \text{PM} \cdot \text{ATO} \quad (4.2)$$

Here, PM denotes the profit generated from 1 DKK of sales and the ATO measures the efficiency in which the Net Operating Assets (NOA) are used to generate sales. The RNOA began strong in 2019, declined through 2021, and then steadily recovered to near initial levels by 2023. As shown in Table 4.2, PM was fairly stable averaging 34.7% with a peak of 36.6% in 2021. The main driver for RNOA changes was ATO fluctuations, which decreased until 2021 and then recovered. This trend indicates underutilized assets and sales challenges until 2021, followed by improved asset efficiency and increased sales. The performance

rebound from 2021 to 2023 is largely due to a surge in sales, especially in the obesity treatment segment.

|      | 2023   | 2022   | 2021   | 2020   | 2019    |
|------|--------|--------|--------|--------|---------|
| RNOA | 88.29% | 73.61% | 67.24% | 79.59% | 100.76% |
| PM   | 34.83% | 36.61% | 33.20% | 33.33% | 35.39%  |
| ATO  | 2.50   | 2.15   | 1.99   | 2.36   | 2.94    |

**Table 4.2:** Decomposition Level 2 - RNOA

**Source:** Novo Nordisk [Annual Report 2018] - [Annual Report 2023] and own calculations

#### 4.1.3 Decomposition Level 3

In the third level decomposition, the focus is on the PM. Table 4.3 shows that COGS relative to sales has decreased since 2021, boosting the gross margin. The EBITDA margin also increased from 2021-2023, driven by lower distribution and administrative costs. However, R&D investment relative to sales has increased since 2019, reflecting a strategic focus on innovation. Despite this, the NOPAT margin declined from 2019 to 2021, then rose through 2023, maintaining overall stability.

|                                     | 2023   | 2022   | 2021   | 2020   | 2019   |
|-------------------------------------|--------|--------|--------|--------|--------|
| Net sales                           | 100.0% | 100.0% | 100.0% | 100.0% | 100.0% |
| Cost of sales                       | -15.4% | -16.1% | -16.8% | -16.5% | -16.1% |
| Gross Margin                        | 84.6%  | 83.9%  | 83.2%  | 83.5%  | 83.5%  |
| Sales and distribution costs        | -24.4% | -26.1% | -26.3% | -25.9% | -26.1% |
| Research and development costs      | -14.1% | -13.6% | -13.2% | -11.7% | -11.8% |
| Administrative costs                | -2.1%  | -2.5%  | -2.9%  | -3.1%  | -3.3%  |
| Other operating income and expenses | 0.1%   | 0.6%   | 0.2%   | 0.4%   | 0.5%   |
| Reversal of D&A                     | 0.1%   | 0.4%   | 0.4%   | 0.4%   | 0.6%   |
| EBITDA Margin                       | 48.2%  | 46.4%  | 45.9%  | 47.2%  | 47.7%  |
| Amortization and depreciation       | -4.1%  | -4.2%  | -4.3%  | -4.5%  | -4.6%  |
| EBIT Margin                         | 44.2%  | 42.3%  | 41.7%  | 42.6%  | 43.0%  |
| Tax of EBIT                         | -8.8%  | -8.4%  | -8.0%  | -8.8%  | -8.6%  |
| Core other operating profit         | 0.0%   | 0.3%   | 0.1%   | 0.0%   | -0.1%  |
| NOPAT Margin                        | 35.3%  | 34.2%  | 33.8%  | 33.8%  | 34.3%  |
| Non-core operating income           | -0.5%  | 2.4%   | -0.6%  | -0.4%  | 1.1%   |
| Total Profit Margin                 | 34.8%  | 36.6%  | 33.2%  | 33.3%  | 35.4%  |

**Table 4.3:** Decomposition Level 3 - PM

**Source:** Novo Nordisk [Annual Report 2018] - [Annual Report 2023] and own calculations

## 5 Budgeting

This section aims to forecast Novo Nordisk's economic performance by integrating strategic and profitability analyses from Sections 3 and 4, along with external market projections.

### 5.1 Forecasting Period

The strategic analysis highlighted Novo Nordisk's growth potential within the diabetes care and most notably the obesity drug sector. Hence, Novo Nordisk is seen as a growth prospect in the future years, which supports a forecasting period of 10 years [Sørensen, 2022].

### 5.2 Revenue

The revenue growth projection will use historical trends and external market forecasts. Novo Nordisk operates in three main segments: diabetes, obesity, and rare diseases. They aim to enter cardiovascular and emerging therapy areas but currently have no approved drugs there. As described in Section 3.2, drug development and approval are highly uncertain. Novo has three drugs in phase 3 trials in this segment [Annual Report 2023], but due to uncertainty, this segment is excluded from the forecast. Therefore, revenue growth estimates focus on the three main segments.

**The diabetes market** is projected to grow at a 6.7% CAGR from 2023-2032 [Yahoo Finance, 2023a]. With an increasing number of diabetes patients and Novo Nordisk's leading market position, the company's market share is rising. Novo's revenue CAGR in this segment was 15.6% from 2019-2023 despite supply-chain challenges. Based on this, an annual growth rate of 15.6% is applied for the first 3 years of the forecast, then it gradually decreases to 6.7%, which continues until the terminal period.

**The obesity market** is rapidly growing, with analysts estimating a CAGR between 5.1-45.6% from 2023-2032 [Statista, 2024a] [Yahoo Finance, 2023b]. Novo Nordisk has grown significantly in this market, with a CAGR of 64.5% from 2019-2023. In 2022, they targeted over 25 billion DKK in obesity sales by 2025, surpassing this with a revenue of 41.6 billion DKK in 2023. This growth is largely driven by the FDA approval of Wegovy in 2021, resulting in a 122.6% CAGR from 2021-2023. Given its importance to Novo's growth, the obesity market's CAGR is influential for the valuation. Novo's revenue from obesity sales is projected to grow at the aggressive analyst estimate of 45.6% gradually stabilizing to the conservative CAGR of 5.1% over a 5-year period. Revenue growth in this market is expected to gradually decline as the market matures and faces potential future price regulations. Furthermore, Novo's patent on Wegovy expires between 2026 and 2032 [Annual Report 2023], depending on the country, which supports a gradual decline in revenue growth.

**The rare disease market** is Novo Nordisk's smallest segment, making up 7.4% of revenue in 2023. The market is projected to grow at an 8.5% CAGR until 2030 [Statista, 2022]. Novo experienced a revenue decline at a -2.7% CAGR from 2019-2023, mainly due to supply constraints. With plans to strengthen their position, this segment is forecasted to gradually increase from a -2.7% CAGR to the market forecasted 8.5% CAGR by 2033.

The split in forecasted revenue growth is summarized in Table 5.1.

### 5.3 EBIT Margin

Novo's EBIT margin has proven stable the last 5 years with a mean of 42.7%. A relatively stable margin during periods of crisis, with the pandemic and the war in Ukraine, as well as periods with high growth within the obesity segment, makes it a reliable metric. Hence, the 5-year average is applied as a constant EBIT-margin for the entire forecast period.

## 5.4 Asset Turnover Rate

Novo Nordisk's ATO indicates how efficiently assets are used to generate revenue. Over the past five years, the ATO varied from 1.99 to 2.94, influenced by market expansion and operational improvements. Future ATO may be affected by Novo's strategic market expansion and production enhancements. While initial investments may lower ATO short-term, they should improve efficiency long-term. Regulatory changes and economic conditions will also impact ATO. Given the uncertainties, a five-year historical average ATO of 2.39 is used as a constant ratio for the forecast period.

## 5.5 Terminal Growth Rate

In the terminal phase, the company is assumed to grow at a constant rate indefinitely. To estimate the terminal growth rate it is assumed that Novo Nordisk will not be able to sustain a growth rate higher than the economy in which it operates [Penman, 2013]. Novo Nordisk is expected to match the economic growth rate of its primary markets, the US and Eurozone. The OECD forecasts real GDP growth of 1.5% in 2060 [OECD, 2023]. Adding the 2% inflation target of the FED and ECB, a terminal growth rate of 3.5% is used for calculating the terminal value of future cash flows.

Tabel 5.1 summarizes the financial projections.

|                             | 2024F | 2025F | 2026F | 2027F | 2028F | 2029F | 2030F | 2031F | 2032F | 2033F | Terminal |
|-----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|
| <b>Revenue Growth</b>       | 19.6% | 14.8% | 10.9% | 7.9%  | 6.8%  | 6.1%  | 6.2%  | 6.2%  | 6.3%  | 6.3%  | 3.5%     |
| <b>EBIT Margin</b>          | 42.7% | 42.7% | 42.7% | 42.7% | 42.7% | 42.7% | 42.7% | 42.7% | 42.7% | 42.7% | 42.7%    |
| <b>Asset Turnover Ratio</b> | 2.39  | 2.39  | 2.39  | 2.39  | 2.39  | 2.39  | 2.39  | 2.39  | 2.39  | 2.39  | 2.39     |
| <b>Revenue Growth Split</b> |       |       |       |       |       |       |       |       |       |       |          |
| <b>Diabetes</b>             | 15.6% | 11.8% | 8.9%  | 6.7%  | 6.7%  | 6.7%  | 6.7%  | 6.7%  | 6.7%  | 6.7%  | 6.70%    |
| <b>Obesity</b>              | 45.6% | 29.4% | 19.0% | 12.2% | 7.9%  | 5.1%  | 5.1%  | 5.1%  | 5.1%  | 5.1%  | 5.10%    |
| <b>Rare diseases</b>        | -2.7% | -1.5% | -0.2% | 1.0%  | 2.3%  | 3.5%  | 4.8%  | 6.0%  | 7.3%  | 8.54% | 8.54%    |

**Table 5.1:** Financial Projections and Revenue Growth Split

**Source:** Own calculations

## 6 Cost of Capital

A company's cost of capital is crucial for valuing it, as it is used for discounting future cash flows. This section estimates Novo Nordisk's weighted average cost of capital (WACC), which combines the returns required by debt and equity holders. WACC represents the expected return for investors considering similar risk alternatives [Koller, 2020]. For a company financed with debt and equity, WACC is defined as follows:

$$WACC = \frac{E}{E + D} \cdot r_E + \frac{D}{E + D} \cdot r_D \cdot (1 - \tau) \quad (6.1)$$

where  $r_E$  and  $r_D$  are the cost of equity and cost of debt and  $\tau$  is the marginal tax rate. The tax rate applied is the corporate tax rate in Denmark of 22%. Equity ( $E$ ) and debt ( $D$ ) are measured using market values. In the following sections, the components of the WACC will be examined.

### 6.1 Cost of Debt

The cost of debt is the return required by the company's creditors. It relies on fluctuations in interest rate levels and the risk of default for the company. As interest expenses are tax deductible, the after-tax-cost of debt is applied to incorporate the interest tax shield into the WACC. Thus, the cost of debt is defined as

$$r_D(1 - \tau) = (r_f + r_s)(1 - \tau) \quad (6.2)$$

where  $r_f$  is the risk-free rate of return and  $r_s$  is the risk premium specific to the company.

#### 6.1.1 Risk-free Rate

The risk-free rate of return is the return on an asset with no risk. In theory, the actual return always equals the expected return. To achieve this, there must be no default risk and no reinvestment risk. Private securities have default risk, so only government bonds can be considered risk-free, since governments control the printing of currency. Additionally, a risk-free investment cannot have intermediate cash flows [Damodaran, 2008]. To satisfy these conditions, a zero-coupon government bond is utilized.

To comply with no intermediate cash flows, the zero-coupon bond's maturity must match the cash flow's time horizon, which would require a recalculation of WACC for each cash flow analysed. Assuming a perpetuity of cash flows, Aswath Damodaran argues that applying a zero-coupon 10-year government bond rate as the risk-free rate on all cash flows is appropriate for mature firms [Damodaran, 2008].

To estimate an appropriate risk-free rate, the currency effect must be considered. Government bond yields vary by country due to currency differences; higher inflation currencies have higher risk-free rates. To mitigate this effect, the risk-free rate should match the currency of the cash flow. Although Novo Nordisk is based in Denmark, 99.2% of its sales are international. Thus, a weighted average of government bonds from relevant countries is used. Table 6.1 shows the yield-to-maturity on 10-year zero-coupon government bonds. Novo doesn't provide a split of net sales by country, so the applied bonds are weighted to represent the US, euro-zone, and Asia.

As shown in Table 6.1, the weighted average bond yield is 3.33%. While these government bonds have minimal default risk, they are not entirely risk-free. To adjust for default risk, a weighted average of the credit default spread is calculated, as seen in Table 6.2, which yields a default risk premium of 0%.

Despite an inadequate country split, the approach helps mitigate currency risk. Spot rates are used instead of historical averages, since the WACC is forward-looking and used to assess future cash flows. Historical averages do not reflect the current economic landscape and could introduce selection bias. Spot rates represent the current market expectations for bond yields over 10 years. After subtracting the

| Country                 | 10-Year Yield | Weight |
|-------------------------|---------------|--------|
| The US                  | 3.93%         | 54.9%  |
| ECB                     | 2.08%         | 21.9%  |
| China                   | 2.55%         | 7.2%   |
| Other                   | n.a.          | 16.0%  |
| <b>Weighted average</b> | <b>3.33%</b>  |        |

**Table 6.1:** Sovereign Bond Yields

**Note:** Yield-to-maturity on 10 year zero-coupon government bonds as of December 29, 2023.

**Source:** [Bloomberg L.P.], [Annual Report 2023]

| Country                 | Rating | CDS Spread   | Weight |
|-------------------------|--------|--------------|--------|
| The US                  | AAA    | 0%           | 54.9%  |
| ECB                     | AAA    | 0%           | 21.9%  |
| China                   | A1     | 0.41%        | 7.2%   |
| Other                   | n.a.   | n.a.         | 16.0%  |
| <b>Weighted average</b> |        | <b>0.00%</b> |        |

**Table 6.2:** Default Risk Premiums

**Note:** Sovereign CDS spreads for each country as at January 1<sup>st</sup>, 2024. CDS applied as proxy for default risk premiums.

**Source:** [Bloomberg L.P.], [Damodaran, 2024b], [ECB, 2024], and own calculations.

average default risk premium, the risk-free rate is 3.33%.

### 6.1.2 Company Specific Risk Premium

In addition to the risk-free rate, investors demand a company-specific risk premium for non-systematic risks. Higher default probability increases the required return, measured by the default spread. For rated firms like Novo Nordisk, the best approach is to use credit ratings from major agencies such as Moody's, Fitch, and S&P to derive the default spread.

Table 6.3 shows the credit rating of Novo Nordisk and a group of peers. Novo Nordisk has a rating of AA- and A1 placing the company between the second and third best investment grade. This aligns with the peer group, with peers rated both slightly better and worse than Novo Nordisk, indicating that a AA-/A1 credit rating is reasonable for estimating the default spread.

| Rating Agency            | S&P | Moody's | Fitch |
|--------------------------|-----|---------|-------|
| Novo Nordisk             | AA- | A1      | n.a.  |
| AstraZenica PLC          | A   | A2      | A-    |
| Sanofi S.A.              | AA  | A1      | n.a.  |
| Novartis AG              | A   | Aa2     | AA-   |
| GlaxoSmithKline PLC      | A   | A2      | n.a.  |
| Roche Holding AG         | AA  | Aa2     | AA    |
| Bayer Aktiengesellschaft | BBB | Baa2    | BBB   |

**Table 6.3:** Credit Ratings

**Note:** Ratings as at December 29, 2023. Peer group based on Bloomberg LP selection.

**Source:** [Bloomberg L.P.]

Based on default spreads published by Aswath Damodaran, a credit rating of AA-/A1 corresponds to a default spread of 0.92% [Damodaran, 2024a]. Inserting each estimated component into Equation (6.2) yields the cost of debt

$$r_D(1 - \tau) = (3.33\% + 0.92\%)(1 - 22\%) = 3.31\% \quad (6.3)$$

The after tax cost of debt is estimated to 3.31%, with the pre-tax rate being 4.25%. The most recent EMTN issued by Novo Nordisk, with 6 years until maturity, was trading at a yield-to-maturity of 2.58% on the valuation date. This rate is reasonably close to the estimated cost of debt, considering the shorter maturity.

## 6.2 Cost of Equity

The cost of equity is the return required by a company's equity investors as compensation for the risk of investing in that company. The cost of equity is typically derived using asset pricing models, such as the Capital Asset Pricing Model (CAPM). According to CAPM, the only non-diversifiable risk is the overall market risk. The risk premium is measured by the market portfolio's systematic risk premium and the investment's sensitivity to this risk, captured by beta. Therefore, CAPM is expressed as follows:

$$r_E = r_f + \beta(E[r_m] - r_f) \quad (6.4)$$

Here  $r_f$  is the risk-free rate of return,  $\beta$  measures the investment's volatility relative to the market and  $E[r_m]$  denotes the expected return of the market portfolio. The term  $E[r_m] - r_f$  represents the market risk premium (MRP). However, CAPM has been criticized for measurement issues and uncertainties with beta and the MRP. Even with statistical significance, beta estimation errors can vary, and the MRP highly depends on the market portfolio definition and varies widely across different studies and textbooks [Penman, 2013]. Alternative models for estimating the cost of equity include the Fama-French 3 Factor model (FF3F) and Arbitrage Pricing Theory (APT), and a discussion of these models is seen in Appendix A.4.1. Considering the scope of the paper, the CAPM offers the most appropriate balance between ease of use and explanatory power for the valuation. Therefore, the subsequent sections will analyze the MRP and beta within the CAPM framework.

### 6.2.1 Market Risk Premium

The Market Risk Premium (MRP) is the difference between the expected market return and the risk-free rate, representing the premium for taking on market risk. There is no consensus on a specific method for estimating MRP, requiring estimates for both the market return and the risk-free rate. The choice of market index is crucial for CAPM reliability; indices from developed, stable, and diversified markets provide more accurate data. Unreliable results occur with indices lacking these characteristics due to large standard errors [Damodaran, 2023]. Therefore, this paper uses the MSCI World Index, covering large and mid-cap stocks in 23 developed countries, providing broad, reliable market exposure.

The risk-free rate should represent a risk-free return aligned with the index used. To account for currency and inflation differences among the safest bonds, an average of U.S. Treasury bonds and a ECB-calculated yield on EUR-area triple AAA rated bonds is used as the risk-free rate. 10-year zero-coupon bonds are applied, and the yields are displayed in Table 6.4.

The ECB yield is lower than US Treasuries, hitting a 12-month low at the end of 2023 due to slower inflation and expected rate cuts [Reuters, 2023b]. To be consistent with the use of spot rates to reflect the current financial environment, the end of December bond yields are used, estimating the risk-free rate at 3.01%.

The next step in estimating the MRP is to determine the expected market return. The three most common methods are the historical MRP, the implied equity risk premium, and the survey MRP [Damodaran, 2023].

| <b>Institution</b>    | <b>Yield</b> |
|-----------------------|--------------|
| FED                   | 3.93%        |
| ECB                   | 2.08%        |
| <b>Risk-free rate</b> | <b>3.01%</b> |

**Table 6.4:** Risk-free rates

**Note:** Yield-to-maturity of 10 year zero-coupon government bonds as of December 29, 2023.  
The risk-free rate is an equally weighted average of the two bond-types.

**Source:** [FRED, 2024], [ECB, 2024]

These methods will be explored and discussed in the following section.

### Historical Market Risk Premium

The historical MRP uses past data to estimate excess return on the index to the risk-free return. Variations arise from choice of the time horizons and averaging method. Longer periods reduce the estimation error, while shorter periods may capture the current investor risk aversion. Additionally, arithmetic averages yield higher premiums than geometric averages but are inconsistent for compounding. Based on different periods and methods, the estimated MRP ranges from 5.74% to 8.48%. For a detailed discussion of this approach, see Appendix A.4.2.

### Implied Equity Risk Premium

The implied equity risk premium (IERP) uses current share prices to estimate the expected market return. Unlike historical methods, IERP is forward-looking, and can be estimated using a discounted dividend (DDM) model. Based on the MSCI World Index share price, dividend yield, and analyst growth estimates, the required return on equity is estimated at 5.79%, resulting in an IERP of 2.79%. However, this underestimates the premium as share buybacks on the index are omitted, which constitute half of the cash flow to equity [Damodaran, 2023]. Due to the lack of share buyback data for the MSCI World Index, an estimate based on S&P 500 is computed and considered more accurate. The resulting market risk premium is 4.74%. For a detailed analysis of this approach, see the Appendix A.4.2.

### Survey Premium

The final method presented is based on survey premiums. Since the MRP represents the premium investors require for equity investments, a logical approach to estimating it is by directly asking investors about their required return on equity. A challenge related to this approach is selecting a representative subset of investors for the aggregate market. In practice, surveys typically concentrate on capturing the expectations of investors, managers, and even academics [Damodaran, 2023]. An extensive survey produced in 2023 involving 15,000 potential participants from 102 countries, including finance and economics professors, analysts, and managers, identifies the market risk premium used among participants for each country [Fernandez et al. 2023]. The survey collected 3,812 answers out of 15,000 possible and reports the results of 80 out of 102 countries, omitting the results for countries with less than 7 answers. Considering the diversity of professions among the participants and the substantial sample size, the survey estimates of the MRP seems reasonable. To maintain consistency with the other methods of estimating the MRP and the subsequent beta calculations, the estimated MRP should mirror the broader market and should not be adjusted to reflect specific characteristics of Novo Nordisk operations. Thus, the estimated MRPs from the survey are displayed in Table 6.5 with country weights equivalent to the market share in the MSCI World Index. The survey MRP is calculated as a weighted average of the country specific MRPs.

Based on survey premiums from Fernandez et al. (2023) the market risk premium is estimated at 5.83%.

| Country            | MRP          | Weight |
|--------------------|--------------|--------|
| The US             | 5.7%         | 70.89% |
| Japan              | 6.1%         | 6.13%  |
| The UK             | 6.0%         | 3.79%  |
| France             | 6.0%         | 3.16%  |
| Canada             | 6.0%         | 3.06%  |
| Other              | 6.3%         | 12.97% |
| <b>Average MRP</b> | <b>5.83%</b> |        |

**Table 6.5:** Survey Market Risk Premium

**Note:** Other is an average of the remaining 19 countries in the MSCI World Index

**Source:** [Fernandez et al. 2023], [MSCI World Index 2024a] and own calculations

### Choosing a Market Risk Premium

The three approaches for determining the MRP propose different estimates ranging from 4.74% to 8.48%. For the valuation of Novo Nordisk, the key is estimating an appropriate risk premium for the future. But which of the methods have the best predictive power? If one believes that historic returns are better at predicting the future than current returns, then the historical approach may be preferred. However, it relies on a strong assumption that investors' risk aversion does not change over time. If one relaxes this assumption by using a shorter and more recent time period for the estimation, it results in unacceptably high standard errors in the estimates. A last downside of this approach is the survivor bias which may lead to overestimation of historical premiums compared to current expected market premiums.

The implied equity risk premium is also subject to criticism. Since it relies on current market data and anticipated future growth, it is highly sensitive to the current economic and geopolitical climate, among other factors. During crises the premium may fluctuate drastically as investors reassess the price of risk. If one believes the aggregate market is significantly under- or overvalued, the historical approach will produce a more accurate premium over long periods. Nonetheless, the current implied premium is found to be the best at predicting actual return premiums for a period of 5-10 years [Damodaran, 2023]. However, as the estimation based on the MSCI World Index significantly underestimates the IERP, the estimate based in S&P Index would have to be applied. The MSCI World Index is chosen as the basis of the CAPM estimation, hence applying the MRP of the S&P Index is inconsistent with the estimation of beta.

To retain consistency within the CAPM framework, it is beneficial to use the survey premium estimate, as it reflects the characteristics of the MSCI World Index. Further, it represents actual current investor expectations, hence avoiding the reliability of historic data similar to the IERP. However, a disadvantage of this approach is that only few practitioners utilize the method, due to the implication of other issues. Empirical findings contradicts the predictive power of survey premiums, documenting a negative relationship between investor sentiment and stock returns [Fisher & Statman 2000]. Nonetheless, the survey premium is estimated to be within the range between the estimated IERP and historical MRP. Thus, maintaining consistency in the model it is found appropriate to use the survey premium, and an MRP of 5.83% is applied.

#### 6.2.2 Beta

Beta measures the systematic risk connected to a particular stock. Hence, it reflects the volatility of a stock's return compared to the return of the overall market [Pratt, 2002]. The most common regression used to estimate beta is the market model

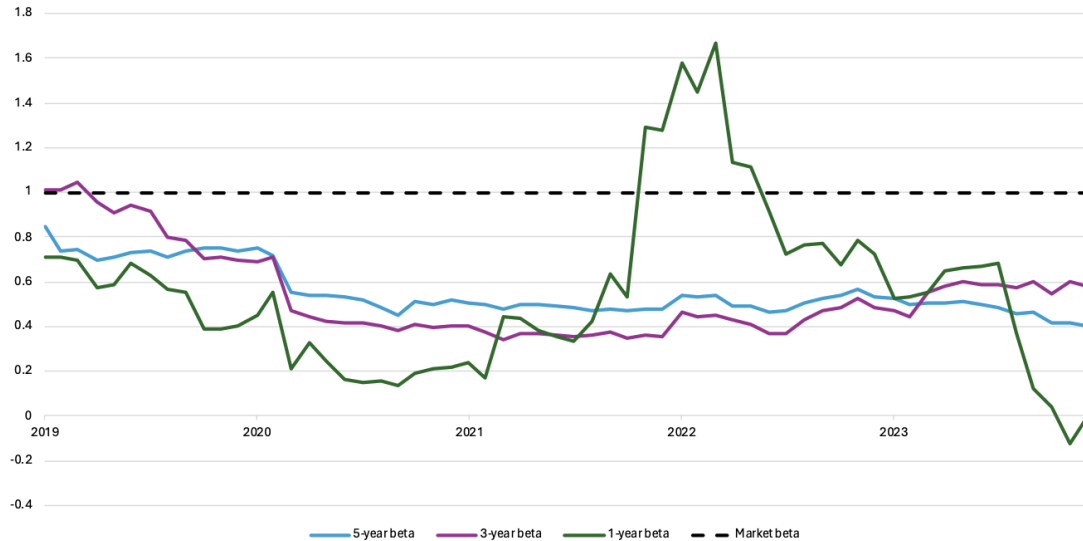
$$r_i = \alpha + \beta r_m + \epsilon \quad (6.5)$$

In this model, the return of a stock ( $r_i$ ) is regressed against the return of the market ( $r_m$ ), with the crucial parameter being the slope  $\beta$ .

When estimating beta two key variables must be considered. First, the time period over which the returns are measured. Second, the frequency of measurements within this time period must be determined [Pratt, 2002]. Commonly, a five-year period of monthly returns is utilized for the estimation. Extending the estimation period can be beneficial as it includes more data points in the regression, potentially leading to a statistically more significant predictor. However, prolonging the sample period may also incorporate outdated company characteristics that no longer reflect the current business environment, which could skew the beta estimation [Damodaran, 1999]. Beta will be estimated based on monthly returns, as using more frequent return periods, such as weekly or daily returns, may lead to systematic biases [Koller, 2020]. From the market model seen in Equation (6.5), beta can be directly computed as follows

$$\beta = \frac{Cov(r_i, r_m)}{Var(r_m)} \quad (6.6)$$

In this case,  $Cov(r_i, r_m)$  is the covariance between the return on Novo Nordisk and the MSCI World Index, and  $Var(r_m)$  is the return variance of the MSCI World Index. Based on 5-year monthly returns, the beta of Novo Nordisk is estimated to 0.41 as at December 29<sup>th</sup>, 2023. As this beta-value is relatively low, rolling betas are estimated from 2019-2023 and illustrated in Figure 6.1 for comparison. The 1-year rolling beta is logically the most volatile, fluctuating between -0.12 and 1.67 during the period. With the 5-year beta being the least volatile series, it seems reasonable to prefer this estimation. It does however decrease from 0.85 to 0.41 indicating a significant change in the stocks sensitivity to overall market fluctuations. While the stock market was significantly impacted by the COVID-19 pandemic and the war in Ukraine during this period, the pharmaceutical sector, as discussed in section 3, was not affected to the same extent. Consequently, this mitigated fluctuations in the return of Novo Nordisk, hence lowering the beta.



**Figure 6.1:** Novo Nordisk Rolling Betas

**Source:** [Bloomberg L.P.], [MSCI World Index 2024a] and own calculations

As Novo's 5-year beta has been relatively volatile, it is evaluated against peer betas. The levered peer betas reported in Table 6.6 are all relatively low and close to the estimated beta of Novo, except the one of Bayer AG. To obtain a more accurate estimate of beta, the beta's are initially unlevered based on each

company's capital structure and relevant corporate tax rate [Damodaran, 1999]. Then, a median industry unlevered beta of 0.38 is used for further calculation, as this adjust for outliers in the rather small sample of peers. The beta value is then relevered based on Novo's capital structure, resulting in a beta of 0.38.

| Peers                                      | 5-year Beta | Debt/Equity | Unlevered Beta |
|--------------------------------------------|-------------|-------------|----------------|
| Novo Nordisk                               | 0.41        | 0.01        | 0.40           |
| AstraZenica PLC                            | 0.19        | 0.14        | 0.17           |
| Sanofi S.A.                                | 0.43        | 0.16        | 0.39           |
| Novartis AG                                | 0.41        | 0.11        | 0.38           |
| GlaxoSmithKline PLC                        | 0.33        | 0.30        | 0.27           |
| Roche Holding AG                           | 0.23        | 0.15        | 0.20           |
| Bayer Aktiengesellschaft                   | 0.98        | 1.36        | 0.46           |
| <b>Median unlevered beta</b>               | <b>0.38</b> |             |                |
| Average unlevered beta                     | 0.32        |             |                |
| <b>Novo Nordisk levered beta by median</b> | <b>0.38</b> |             |                |
| Novo Nordisk levered beta by average       | 0.33        |             |                |

**Table 6.6:** Peer Analysis of Industry Betas

**Note:** All estimates are based on data as at the valuation date and estimated against MSCI World Index.

**Source:** [Bloomberg L.P.] and own calculations

To mitigate the effect of the volatility in company betas indicated in Figure 6.1, an additional adjustment is made. Several studies argue that company betas converge towards one over time, as surviving firms tend to grow, diversify, and accumulate more assets, which drives their beta towards one [Kolb, 1989]. The adjusted beta is simply calculated as

$$\beta_{adj} = \frac{2}{3}0.38 + \frac{1}{3}1 = 0.59 \quad (6.7)$$

Hence, a beta of 0.59 is applied in the analysis.

### Cost of Equity

Inserting the estimated value of the parameters in the CAPM yields the following cost of equity

$$r_E = 3.01\% + 0.59 \cdot 5.83\% = 6.43\% \quad (6.8)$$

### 6.2.3 WACC

The WACC is computed based on the cost of equity and debt derived in the preceding sections. The market value of equity is derived as the product of outstanding stocks and share price at valuation date, and the market value of debt is proxied by the net financial obligations reported by Novo [Penman, 2013]. 4,483 million outstanding shares at a price of 698.1 DKK yields an equity value of 3,129,443 mDKK, while the value of debt is 28,278 mDKK.

$$WACC = \frac{3,129,443}{3,157,721} \cdot 6.43\% + \frac{28,278}{3,157,721} \cdot 3.31\% = 6.40\% \quad (6.9)$$

Inserting this into Equation (6.1) yields a WACC of 6.40%, which will be applied in the valuation.

## 7 Valuation

In this section, the intrinsic value of Novo Nordisk is calculated using the discounted cash flow (DCF) model, and the reliability of the estimated value is further investigated in a sensitivity analysis.

### 7.1 Discounted Cash Flow Model

Utilizing the DCF model, the enterprise value of a firm is determined as the present value of the forecasted future free cash flows to the firm (FCFF). The enterprise value is computed as follows

$$EV_0 = \sum_{t=1}^n \frac{FCFF_t}{(1+WACC)^t} + \frac{FCFF_{n+1}}{(WACC-g)} \cdot \frac{1}{(1+WACC)^n} \quad (7.1)$$

Here,  $EV_0$  is the enterprise value at valuation date,  $FCFF_t$  is the free cash flow to firm in period  $t$ , and  $n$  denotes the number of forecasted periods. The term  $\frac{FCFF_{n+1}}{(WACC-g)}$  represents the terminal value, where  $g$  is a perpetual growth rate given by the terminal growth rate. Hence, the EV is the present value of free cash flows forecasted for the next 10 years, plus the terminal value representing the perpetually growing cash flows beyond this period. The market value of equity is then obtained by subtracting the market value of the debt from the EV.

The DCF valuation of Novo Nordisk is seen in Figure 7.1. The estimated value per share is DKK 899, which is 28.8% above the actual share price of DKK 698, indicating that the stock is undervalued.

| In DKK millions                  | 2023A            | 2024F   | 2025F   | 2026F   | 2027F   | 2028F   | 2029F   | 2030F   | 2031F   | 2032F   | 2033F   | Terminal  |
|----------------------------------|------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|-----------|
| Revenue                          | 232,261          | 277,823 | 319,003 | 353,756 | 381,707 | 407,809 | 432,785 | 459,544 | 488,213 | 518,937 | 551,884 | 571,200   |
| EBIT                             | 102,574          | 118,763 | 136,366 | 151,223 | 163,171 | 174,329 | 185,006 | 196,444 | 208,699 | 221,833 | 235,917 | 244,175   |
| Operating Income                 | 82,055           | 92,635  | 106,366 | 117,954 | 127,273 | 135,976 | 144,304 | 153,226 | 162,786 | 173,030 | 184,016 | 190,456   |
| Turnover rate (ATO)              | 2.50             | 2.39    | 2.39    | 2.39    | 2.39    | 2.39    | 2.39    | 2.39    | 2.39    | 2.39    | 2.39    | 2.39      |
| 1/ATO                            | 0.40             | 0.42    | 0.42    | 0.42    | 0.42    | 0.42    | 0.42    | 0.42    | 0.42    | 0.42    | 0.42    | 0.42      |
| Net operating assets             | 101,012          | 116,357 | 133,604 | 148,159 | 159,865 | 170,798 | 181,258 | 192,465 | 204,472 | 217,340 | 231,139 | 239,229   |
| Free cash flows                  |                  | 77,290  | 89,119  | 103,398 | 115,567 | 125,044 | 133,844 | 142,020 | 150,779 | 160,162 | 170,217 | 182,366   |
| Discount factor                  |                  | 1.06    | 1.13    | 1.20    | 1.28    | 1.36    | 1.45    | 1.54    | 1.64    | 1.75    | 1.86    | 1.98      |
| Present value of FCF             |                  | 72,642  | 78,723  | 85,844  | 90,177  | 91,705  | 92,255  | 92,004  | 91,805  | 91,654  | 91,550  |           |
| Terminal value of FCF            |                  |         |         |         |         |         |         |         |         |         |         | 6,292,371 |
| NPV of FCF                       | 878,359          |         |         |         |         |         |         |         |         |         |         |           |
| NPV of terminal value            | 3,180,802        |         |         |         |         |         |         |         |         |         |         |           |
| Enterprise value                 | 4,059,162        |         |         |         |         |         |         |         |         |         |         |           |
| Net financial obligations        | 28,278           |         |         |         |         |         |         |         |         |         |         |           |
| <b>Equity value</b>              | <b>4,030,884</b> |         |         |         |         |         |         |         |         |         |         |           |
| # of shares                      | 4,483            |         |         |         |         |         |         |         |         |         |         |           |
| <b>Estimated value per share</b> | <b>899</b>       |         |         |         |         |         |         |         |         |         |         |           |
| Share price at 31/12/2023        | 698              |         |         |         |         |         |         |         |         |         |         |           |
| Difference                       | 28.8%            |         |         |         |         |         |         |         |         |         |         |           |

**Figure 7.1: DCF Valuation**

**Source:** [Annual Report 2018]-[Annual Report 2023] and own calculations.

To provide a basis for comparison, estimated target prices and recommendations from various financial institutes are presented in Table 7.1. The institutes all estimate a share price exceeding the actual price, and they all agree on a "buy" recommendation, except for Jyske Bank advising investors to "hold". The average target price is DKK 817, which is 17% above the actual price. This estimate is lower than the valuation of DKK 899 per share, but it does however indicate that the share may in fact be undervalued. The analyst-estimates vary by DKK 105 per share, resulting in a spread of DKK 471 billion in the estimated market value of equity. Assuming similarity in the applied model framework of the valuations, this highlights the importance of analyzing the sensitivity of the valuation to its assumptions and parameters.

| Institute                | Date       | Target     | Recommendation |
|--------------------------|------------|------------|----------------|
| AlphaValue/Baader Europe | 01.01.2024 | 825        | Buy            |
| Jyske Bank               | 29.12.2023 | 745        | Hold           |
| Barclays                 | 22.12.2023 | 825        | Buy            |
| Nordea Bank              | 19.12.2023 | 792        | Buy            |
| HSBC                     | 18.12.2023 | 850        | Buy            |
| SEB Bank                 | 08.12.2023 | 835        | Buy            |
| JP Morgan                | 05.12.2023 | 850        | Buy            |
| <b>Average Price</b>     |            | <b>817</b> |                |

**Table 7.1:** Analyst Recommendations

**Source:** [Bloomberg L.P.]

## 7.2 Sensitivity Analysis

As the focus of this paper has been on the Cost of Capital, this sensitivity analysis will primarily investigate how variations in parameters within this area affect the valuation. The WACC rate is pivotal in the result of the DCF-valuation, and variations in this rate has significant impact on the intrinsic value. As seen in Figure 7.2 the estimated share prices varies from DKK 736 to DKK 1,148 when the WACC rate is 7% and 5.8% respectively. The figure illustrates how changes in the WACC rate of 0.2 pp can lead to changes in the share price of close to DKK 100. This clearly indicates a high sensitivity to the estimated discount rate. The NPV of the terminal value accounts for the greater share of the enterprise value, hence variations in the terminal growth rate are also examined in the figure. The figure shows how a decrease or increase in the growth rate of 0.6 pp results in a share price of DKK 778 or DKK 1,083 respectively.

|      |       | Terminal Growth Rate |       |       |       |       |       |       |
|------|-------|----------------------|-------|-------|-------|-------|-------|-------|
| WACC |       | 2.90%                | 3.10% | 3.30% | 3.50% | 3.70% | 3.90% | 4.10% |
|      | 5.80% | 952                  | 1,008 | 1,072 | 1,148 | 1,239 | 1,348 | 1,483 |
|      | 6.00% | 887                  | 934   | 989   | 1,052 | 1,126 | 1,214 | 1,321 |
|      | 6.20% | 829                  | 870   | 917   | 970   | 1,032 | 1,104 | 1,191 |
|      | 6.40% | 778                  | 814   | 854   | 899   | 951   | 1,012 | 1,083 |
|      | 6.60% | 733                  | 764   | 799   | 838   | 882   | 933   | 993   |
|      | 6.80% | 693                  | 720   | 750   | 784   | 822   | 866   | 916   |
|      | 7.00% | 656                  | 680   | 706   | 736   | 769   | 807   | 849   |

**Figure 7.2:** Sensitivity Analysis: WACC & Terminal Growth Rate

**Source:** Own calculations

Diving deeper into the parameters of the WACC, Figure 7.3 shows variations in two parameters of the CAPM estimation: the beta and MRP. In the calculation of Novo's WACC the cost of equity is weighted at 99%, hence evaluating changes in the parameter values of CAPM is essential. In the analysis of the 5-year rolling beta of Novo in Section 6.2.2, the beta value was seen to decrease from 2019-2023, which was also the case for the majority of the peers [Bloomberg L.P.]. The 3-year beta has however been increasing since 2021, hence the 5-year beta is set to increase if the last years development continues. Figure 7.3 shows that an increase in beta of 0.15 alone yields a share price of DKK 680, which would change the conclusion to the share being overvalued.

Several methods for estimating the MRP was analysed and discussed in Section 6.2.1. Table 7.3 illustrates that the MRP has significant impact on the valuation as well. If the rate was to increase or decrease with 0.6 pp the price per share would vary by DKK 233. An interesting result from the figure is that an increase of 0.1 in the value of beta and 0.4 pp in the MRP yields an estimated share price of DKK 684,

which is slightly below the actual price at valuation date. This highlights that the estimated share price of DKK 899 must be interpreted with great caution as it is highly sensitive to minor changes in the values of the parameters.

|      |      | Market Risk Premium |       |       |       |       |       |       |
|------|------|---------------------|-------|-------|-------|-------|-------|-------|
|      |      | 5.23%               | 5.43% | 5.63% | 5.83% | 6.03% | 6.23% | 6.43% |
| Beta | 0.44 | 1,507               | 1,434 | 1,368 | 1,307 | 1,251 | 1,200 | 1,152 |
|      | 0.49 | 1,307               | 1,245 | 1,189 | 1,137 | 1,089 | 1,045 | 1,005 |
|      | 0.54 | 1,152               | 1,099 | 1,050 | 1,005 | 963   | 925   | 889   |
|      | 0.59 | 1,030               | 982   | 939   | 899   | 862   | 829   | 797   |
|      | 0.64 | 930                 | 887   | 849   | 813   | 780   | 750   | 721   |
|      | 0.69 | 846                 | 808   | 773   | 741   | 711   | 684   | 658   |
|      | 0.74 | 776                 | 742   | 710   | 680   | 653   | 628   | 605   |

**Figure 7.3:** Sensitivity Analysis: Market Risk Premium & Beta

**Source:** Own calculations

## 8 Discussion & Conclusion

In this paper, an estimate of the intrinsic value of Novo Nordisk A/S is derived using a DCF model, resulting in a share price of DKK 899 as at December 31, 2023. The price suggests that Novo Nordisk was undervalued compared to the actual price of DKK 698 at the date of valuation. This paper primarily focus on the cost of capital of Novo Nordisk, which is estimated to 6.40%, with a cost of equity of 6.43% and a cost of debt of 3.31%. Even though the estimations are based on extensive research and determined with a theoretical foundation, they are subject to a considerable uncertainty, due to the many assumptions, methodological and theoretical choices. Therefore, the conclusion that Novo Nordisk was undervalued should be interpreted with caution and should not be the sole factor in evaluating the stock's investment potential. The valuation is, nonetheless, supported by an average target price of DKK 817 from financial institutions, accompanied by a collective buy recommendation. Evaluating the estimated share price against the actual development five months after the valuation date suggests that the valuation may have some reliability. The share price has steadily increased since the valuation date, and at May 31, 2024 the closing price was DKK 927.30.

Since Novo Nordisk is almost entirely equity-financed, a significant part of the paper focuses on the cost of equity. A valuable insight from the sensitivity analysis is that variations in the WACC rate has significant impact on the final valuation. Specifically, even minor changes in the values of beta and the market risk premium can significantly alter the conclusion of the paper. This paper found the survey MRP as the most appropriate measure of the MRP, but it can reasonably be argued that either the historical MRP or the IERP should have been applied. Ultimately this would either decrease or increase the intrinsic value of Novo Nordisk showcasing the considerable uncertainty despite the theoretic foundation of the estimation.

Novo Nordisk is seen as a growth prospect with great possibility of expansion within the segment of obesity drugs. This paper omits an analysis of variations in the projections of e.g. revenue growth, due to the emphasis on the cost of capital and the limited length of the paper. However, analysing different growth scenarios, such as a bull and bear case, and the implications of this on the valuation is highly relevant for further studies.

In practice asset management firms perform valuations based on various other methods. Here, the analysis of multiples is widely used, due to its simplicity and efficiency in the valuation of companies. A valuation based on the DCF model is time-consuming and connected to numerous assumptions, whereas the analysis

of multiples has fewer inputs. Hence, it would be valuable to extend this paper with an analysis of multiples for comparison to the current valuation.

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# A Appendix

## A.1 PESTEL Analysis

In this section, the PESTEL framework is used to identify key external factors impacting Novo’s performance.

### Political

Novo Nordisk operates within a complex political landscape that significantly influences its business operations. Political factors affecting Novo Nordisk include healthcare regulations, government funding policies, pricing scrutiny, and international trade policies.

The company’s operations are subject to the healthcare policies of various countries, which influence market access and pricing strategies. For instance, in the United States and Europe, healthcare regulations determine how quickly Novo Nordisk can bring new drugs to the market and at what price. These regulations vary significantly between regions, affecting overall profitability. In recent years, there has been increasing political pressure on pharmaceutical pricing, particularly in the United States. The increased political scrutiny of drug pricing is exemplified by Senator Bernie Sanders’ criticism of the pricing strategies for insulin and obesity drugs like Ozempic and Wegovy in the U.S., suggesting a potential for more stringent future regulations on drug prices [MedWatch, 2023].

Looking forward, the political environment is expected to have a mixed influence on Novo Nordisk. While increased scrutiny over drug pricing could pose challenges by potentially capping prices and squeezing profit margins, supportive policies around healthcare innovation and chronic disease management could benefit the company. Additionally, Novo Nordisk’s extensive lobbying efforts and strategic partnerships with healthcare stakeholders may help mitigate negative impacts and leverage political support for favorable policies.

### Economic

The global economic development affects Novo Nordisk, with currency fluctuations impacting earnings given the company’s global presence. Economic downturns can variably affect healthcare spending, although prescription drugs see more stable demand compared to other consumer goods. Novo Nordisk’s recent performance shows robust revenue growth, despite global economic uncertainties, partly driven by strong demand for its diabetes and obesity treatments. The company recognizes foreign exchange rate volatility as one of its primary financial risks, especially under the current market conditions [Annual Report 2023]. Any changes in key currency exchange rates can have a direct and substantial effect on the company’s cash flows, posing a risk to its profitability. While most of Novo Nordisk’s sales are conducted in major currencies such as USD, EUR, CNY, JPY, and CAD, the risk associated with EUR-flows is considered low due to Denmark’s fixed currency policy towards the euro.

Novo Nordisk’s significant exposure to the USD is attributed to the substantial size of the U.S. market making up 55% of total sales. To mitigate the risks associated with foreign exchange fluctuations, Novo Nordisk actively hedges parts of its assets, liabilities, and cash flows to minimize the impact of volatility in crucial currencies [Annual Report 2023]. The overall risk posed by exchange rate fluctuations to Novo Nordisk’s future value is considered to be between low and medium.

### Social

Demographic trends such as an aging population and increasing rates of obesity and diabetes worldwide expand the potential market for Novo Nordisk. Social factors also include changing perceptions towards obesity as a treatable condition, which can drive demand for related pharmaceutical treatments. As the company’s profitability is heavily dependent on managing lifestyle-related diseases, any changes in these health and lifestyle trends can have a direct impact on the size of Novo Nordisk’s market and its customer

base.

The global trend indicates a projected increase in people with diabetes of 19.7%, rising from 537 million individuals in 2021 to an estimated 643 million by 2030 [IDF, 2021]. By 2045, the forecast estimates a further increase to a total of 783 million people living with diabetes. Similarly, the worldwide trend of overweight and obese people indicates an increase from 2.6 billion to 4.0 billion from 2020 to 2035, with the proportion of overweight and obese people relative to the total population increasing from 38% to 51% [World Obesity, 2023].

With a significant increase in potential customer base for the products sold by Novo Nordisk, the social impact is assessed to be positive on future profitability of Novo Nordisk.

### **Technological**

Advancements in biotechnology and healthcare are critical for Novo Nordisk. Innovations in drug delivery systems, diabetes care devices, and treatment methodologies directly influence the company's product portfolio and its effectiveness. Staying ahead in technology adoption is crucial for maintaining competitive advantage and responding to the healthcare needs efficiently. In 2022, Novo Nordisk entered strategic partnership with Microsoft aiming to utilize advancements in artificial intelligence and big data. The goal of the collaboration is to enhance and accelerate the R&D process. The partnership is expected to lead to faster and upscaled discovery of new products, thereby increasing Novo Nordisk's innovative capacity and long-term competitive edge. Hence, the technological segment may have a positive effect on the future profitability of the company.

### **Environmental**

In response to the ongoing climate crisis, corporations across all industries are actively seeking to reduce their carbon footprints and incorporate Environmental, Social, and Governance (ESG) practices throughout their entire value chains. In recent years, a growing demand for ESG-friendly investments has led to a significant shift of capital towards sustainable investments [Statista, 2024]. Consequently, ESG practices are increasingly becoming a foundational element of corporate governance. A lack of focus on ESG can deter both large institutional and private investors, potentially reducing a company's market value. Novo Nordisk has set a target reaching net zero emissions across the entire value chain by 2045. Currently, the company has an ESG risk rating of 'Medium', given one of the leading ESG rating firms Morningstar Sustainalytics, which places Novo among the top 12% within the industry of pharmaceuticals [Novo Nordisk, 2024]. A relatively strong position in relation to ESG measures within the industry mitigates the threat of withdrawal from investors due to ESG concerns, which is positive for Novo Nordisk's future value.

### **Legal**

Intellectual property rights, compliance with international health regulations, and patent laws shape Novo Nordisk's business strategy. The legal conditions dictate how Novo Nordisk conducts clinical trials, protects its innovations, and competes in the global market. Hence, Novo Nordisk heavily relies its drugs being approved by various governmental institutions, that regulate and review the pharmaceutical industry. Developing a new pharmaceutical drug typically requires 10-15 years, with an average cost of USD 2.6 billion. This extensive time and financial investment reflect the challenges and rigorous testing involved. Additionally, only 12% of drugs that enter clinical trials ultimately gain FDA approval, highlighting the stringent standards and high risk associated with drug development. This highlights the significant impact of patent laws on Novo Nordisk's profitability as these laws grant temporary monopolies on their drugs, allowing the company to exclusively market their innovations and recover R&D costs. Upon patent expiration, Novo Nordisk faces competition from generic manufacturers, which reduces prices and profits for the original drug. Thus, the duration and strength of patent protection are critical to maintaining

Novo Nordisk's financial health and capacity to invest in new drug development.

## **A.2 Porter's Five Forces**

This section uses Porter's Five Forces to examine key industry trends affecting Novo's competitive environment.

### **Threat of New Entrants: Low**

When evaluating the threat of new entrants in the pharmaceutical industry, the analysis concentrates on the barriers that new companies encounter. High barriers to entry, such as substantial initial investment and regulatory requirements, tend to limit market access, resulting in fewer competitors. This scarcity of competitors can lead to higher profits due to limited pricing competition. Conversely, low barriers increase competition and thus reduce profitability [Sørensen, 2022].

In the pharmaceutical sector, entry barriers are particularly high due to the extensive capital required for production facilities and laboratory equipment. Additionally, as described in Section 3.1, the comprehensive and lengthy R&D process from idea to approved drug is connected to a great cost. If a company succeeds in obtaining regulatory-approval of a drug, the company is granted a monopoly on the product. In the US, the law ensures 20 years of patent [FDA, 2015]. However, as 10-15 years are spent on R&D, which may be included in the lifespan of the patent, only a limited time is left for the company to break-even on the original investment. When the patent expires, generic variants of the drug will enter the market, lowering the price of the original drug. The variants are often produced by established and large competitors, that may benefit from economies of scale. Thus, the barriers of entry to the industry are high, reducing the risk of new entrants.

### **Bargaining Power of Suppliers: Low**

If the availability of suppliers is limited and they provide critical resources with few substituting products, the bargaining power of suppliers is considered high [Sørensen, 2022]. While Novo Nordisk may rely on specific suppliers for certain specialized components or materials, its global presence and ability to switch suppliers, coupled with its extensive network of over 60,000 suppliers, help mitigate this risk [Annual Report 2023].

### **Bargaining Power of Buyers: Low-to-moderate**

The bargaining power of buyers in Novo Nordisk's markets varies. In individual consumer markets, patients have limited bargaining power due to the essential nature of Novo Nordisk's products. However, large-scale buyers like government health programs and insurance companies in countries like the U.S. wield some power due to their large purchasing volumes and ability to negotiate lower prices, which can pressure pricing and profitability. Novo Nordisk specializes in manufacturing medications for chronic and lifestyle diseases such as diabetes and obesity, which are crucial for the patients' health and well-being. These drugs are not easily substituted due to their specific therapeutic effects and the serious health implications of the conditions they treat. Consequently, this necessity reduces the price sensitivity of the buyers, as patients depend heavily on these drugs for their quality of life. Therefore, despite the potential for large buyers like insurance companies and government health programs to negotiate prices, the critical nature of Novo Nordisk's medications inherently limits the overall bargaining power of buyers. Thus, the bargaining power of buyers is considered low-to-moderate.

### **Threat of Substitute Products: Moderate-to-high**

There is a moderate to high threat of substitutes in the pharmaceutical industry. This includes not only generic and biosimilar drugs competing against the original drug once patent protection expires but also alternative therapies such as new drugs with different active ingredients or therapeutic methods like

medical devices and lifestyle interventions. As all patents will expire with time, the generic drugs will pose a threat as substitutes to Novo Nordisk's original medication within all segments. However, Novo Nordisk maintains a competitive edge through continuous innovation and by expanding indications for existing drugs, which helps to lengthen the product lifecycle and fend off competition.

### **Competitive Rivalry: Moderate**

In the pharmaceutical sector, especially in the diabetes and obesity medicine markets, competition is limited to few large companies due to the industry's extensive use of patents. These patents effectively create monopoly-like conditions for their products, limiting market rivalry by protecting their innovations and controlling market entry for extended periods. Novo currently has 6 patents within diabetes care, a patent for the obesity drug Wegovy, and 2 in the segment of rare diseases [Annual Report 2023]. The biggest competitors in the diabetes care segment are Sanofi and Eli Lilly, with the three companies constituting more than 90% of the global insulin market share [Knox, 2020]. The rivalry is also present in the obesity segment and is set against a backdrop of increasing sales projections for obesity drugs, indicating a robust market potential that could reach as high as USD 77 billion by 2030 [Morgan Stanley, 2023]. Novo Nordisk and its major competitors are positioned to benefit significantly from this growth, provided they continue to innovate and efficiently manage market challenges such as supply chain issues and regulatory hurdles. As highlighted in the PESTEL analysis, the expanding market size for obesity and diabetes patients is contributing to a reduction in industry rivalry by enlarging the total market. The growth in potential customers dilutes the competitive intensity among firms by increasing the available opportunities for market share expansion. Based on these observations, the level of industry rivalry is assessed to be moderate.

## A.3 Financial Statements

| <b>Balance sheet</b>                 |      |                |                |                |                |                |                |
|--------------------------------------|------|----------------|----------------|----------------|----------------|----------------|----------------|
| at 31 December                       |      |                |                |                |                |                |                |
| DKK million                          |      |                |                |                |                |                |                |
|                                      | Note | 2023           | 2022           | 2021           | 2020           | 2019           | 2018           |
| <b>Assets</b>                        |      |                |                |                |                |                |                |
| Intangible assets                    | 3.1  | 60,406         | 50,939         | 43,171         | 20,657         | 5,835          | 5,145          |
| Property, plant and equipment        | 3.2  | 90,961         | 66,671         | 55,362         | 50,269         | 50,551         | 41,891         |
| Investments in associated companies  |      | 410            | 327            | 525            | 582            | 474            | 531            |
| Deferred income tax assets           | 2.6  | 20,380         | 13,904         | 8,672          | 5,865          | 4,121          | 2,893          |
| Other receivables and prepayments    |      | 1,430          | 206            | 267            | 674            | 841            | 0              |
| Other financial assets               |      | 1,253          | 1,016          | 916            | 1,066          | 1,334          | 1,242          |
| <b>Total non-current assets</b>      |      | <b>174,840</b> | <b>133,063</b> | <b>108,913</b> | <b>79,113</b>  | <b>63,156</b>  | <b>51,702</b>  |
| Inventories                          | 3.3  | 31,811         | 24,388         | 19,621         | 18,536         | 17,641         | 16,336         |
| Trade receivables                    | 3.4  | 64,770         | 50,560         | 40,643         | 27,734         | 24,912         | 22,786         |
| Tax receivables                      |      | 2,423          | 940            | 1,119          | 289            | 806            | 1,013          |
| Other receivables and prepayments    |      | 8,068          | 6,005          | 5,037          | 4,161          | 3,434          | 3,090          |
| Marketable securities                | 4.4  | 15,838         | 10,921         | 6,765          | 0              | 0              | 204            |
| Derivative financial instruments     | 4.5  | 2,344          | 2,727          | 1,690          | 2,332          | 188            | 0              |
| Cash at bank                         | 4.7  | 14,392         | 12,653         | 10,720         | 12,757         | 15,475         | 15,638         |
| <b>Total current assets</b>          |      | <b>139,646</b> | <b>108,194</b> | <b>85,595</b>  | <b>65,809</b>  | <b>62,456</b>  | <b>59,067</b>  |
| <b>Total assets</b>                  |      | <b>314,486</b> | <b>241,257</b> | <b>194,508</b> | <b>144,922</b> | <b>125,612</b> | <b>110,769</b> |
|                                      |      |                |                |                |                |                |                |
|                                      |      |                |                |                |                |                |                |
| DKK million                          | Note | 2023           | 2022           | 2021           | 2020           | 2019           | 2018           |
| <b>Equity and liabilities</b>        |      |                |                |                |                |                |                |
| Share capital                        | 4.3  | 451            | 456            | 462            | 470            | 480            | 490            |
| Treasury shares                      | 4.3  | -5             | -6             | -6             | -8             | -10            | -11            |
| Retained earnings                    |      | 104,839        | 80,587         | 72,004         | 63,774         | 57,817         | 53,406         |
| Other reserves                       | 4.3  | 1,276          | 2,449          | -1,714         | -911           | -694           | -2,046         |
| <b>Total equity</b>                  |      | <b>106,561</b> | <b>83,486</b>  | <b>70,746</b>  | <b>63,325</b>  | <b>57,593</b>  | <b>51,839</b>  |
| Borrowings                           | 4.6  | 20,528         | 24,318         | 12,961         | 2,897          | 3,009          | 0              |
| Deferred income tax liabilities      | 2.6  | 10,162         | 7,061          | 5,271          | 2,502          | 80             | 118            |
| Retirement benefit obligations       |      | 742            | 762            | 1,280          | 1,399          | 1,334          | 1,256          |
| Other liabilities                    |      | 189            | 100            | 360            | 0              | 0              | 0              |
| Provisions                           | 3.5  | 6,649          | 4,590          | 4,374          | 4,526          | 4,613          | 3,392          |
| <b>Total non-current liabilities</b> |      | <b>38,270</b>  | <b>36,831</b>  | <b>24,246</b>  | <b>11,324</b>  | <b>9,036</b>   | <b>4,766</b>   |
| Borrowings                           | 4.6  | 6,478          | 1,466          | 13,684         | 7,459          | 1,474          | 515            |
| Trade payables                       |      | 25,606         | 15,587         | 8,870          | 5,717          | 6,358          | 6,756          |
| Tax payables                         |      | 7,116          | 7,091          | 3,658          | 3,913          | 4,212          | 4,610          |
| Other liabilities                    |      | 28,705         | 23,606         | 19,600         | 14,005         | 15,085         | 14,098         |
| Derivative financial instruments     | 4.5  | 1,272          | 2,903          | 2,184          | 1,365          | 734            | 2,024          |
| Provisions                           | 3.5  | 100,478        | 70,287         | 51,520         | 34,814         | 31,120         | 26,161         |
| <b>Total current liabilities</b>     |      | <b>169,655</b> | <b>120,940</b> | <b>99,516</b>  | <b>70,273</b>  | <b>58,983</b>  | <b>54,164</b>  |
| <b>Total liabilities</b>             |      | <b>207,925</b> | <b>157,771</b> | <b>123,762</b> | <b>81,597</b>  | <b>68,019</b>  | <b>58,930</b>  |
| <b>Total equity and liabilities</b>  |      | <b>314,486</b> | <b>241,257</b> | <b>194,508</b> | <b>144,922</b> | <b>125,612</b> | <b>110,769</b> |

Figure A.1: Reported Balance Sheet

Note: All values are in DKK million

Source: [Annual Report 2018]-[Annual Report 2023]

|                                               | Note | 2023           | 2022           | 2021           | 2020           | 2019           | 2018           |
|-----------------------------------------------|------|----------------|----------------|----------------|----------------|----------------|----------------|
| <b>Operating assets</b>                       |      |                |                |                |                |                |                |
| Intangible assets                             | 3.1  | 60,406         | 50,939         | 43,171         | 20,657         | 5,835          | 5,145          |
| Property, plant and equipment                 | 3.2  | 90,961         | 66,671         | 55,362         | 50,269         | 50,551         | 41,891         |
| Investments in associated companies           |      | 410            | 327            | 525            | 582            | 474            | 531            |
| Deferred income tax assets                    | 2.6  | 20,380         | 13,904         | 8,672          | 5,865          | 4,121          | 2,893          |
| Other receivables and prepayments             |      | 1,430          | 206            | 267            | 674            | 841            | 0              |
| Inventories                                   | 3.3  | 31,811         | 24,388         | 19,621         | 18,536         | 17,641         | 16,336         |
| Trade receivables                             | 3.4  | 64,770         | 50,560         | 40,643         | 27,734         | 24,912         | 22,786         |
| Tax receivables                               |      | 2,423          | 940            | 1119           | 289            | 806            | 1013           |
| Other receivables and prepayments             |      | 8,068          | 6,005          | 5,037          | 4,161          | 3,434          | 3,090          |
| <b>Operating assets</b>                       |      | <b>280,659</b> | <b>213,940</b> | <b>174,417</b> | <b>128,767</b> | <b>108,615</b> | <b>93,685</b>  |
| <b>Operating liabilities</b>                  |      |                |                |                |                |                |                |
| Deferred income tax liabilities (non-current) |      | 10,162         | 7,061          | 5,271          | 2,502          | 80             | 118            |
| Retirement benefit obligations (non-current)  |      | 742            | 762            | 1,280          | 1,399          | 1,334          | 1,256          |
| Other liabilities (non-current)               |      | 189            | 100            | 360            | 0              | 0              | 0              |
| Provisions (non-current)                      | 3.5  | 6,649          | 4,590          | 4,374          | 4,526          | 4,613          | 3,392          |
| Trade payables (current)                      |      | 25,606         | 15,587         | 8,870          | 5,717          | 6,358          | 6,756          |
| Tax payables (current)                        |      | 7,116          | 7,091          | 3,658          | 3,913          | 4,212          | 4,610          |
| Other liabilities (current)                   |      | 28,705         | 23,606         | 19,600         | 14,005         | 15,085         | 14,098         |
| Provisions (current)                          | 3.5  | 100,478        | 70,287         | 51,520         | 34,814         | 31,120         | 26,161         |
| <b>Operating liabilities</b>                  |      | <b>179,647</b> | <b>129,084</b> | <b>94,933</b>  | <b>66,876</b>  | <b>62,802</b>  | <b>56,391</b>  |
| <b>Net Operating Assets (NOA)</b>             |      | <b>101,012</b> | <b>84,856</b>  | <b>79,484</b>  | <b>61,891</b>  | <b>45,813</b>  | <b>37,294</b>  |
| <b>Financial assets</b>                       |      |                |                |                |                |                |                |
| Other financial assets                        |      | 1,253          | 1,016          | 916            | 1,066          | 1,334          | 1,242          |
| Marketable securities                         | 4.4  | 15,838         | 10,921         | 6,765          | 0              | 0              | 204            |
| Derivative financial instruments              | 4.5  | 2,344          | 2,727          | 1,690          | 2,332          | 188            | 0              |
| Cash at bank                                  | 4.7  | 14,392         | 12,653         | 10,720         | 12,757         | 15,475         | 15,638         |
| <b>Financial assets</b>                       |      | <b>33,827</b>  | <b>27,317</b>  | <b>20,091</b>  | <b>16,155</b>  | <b>16,997</b>  | <b>17,084</b>  |
| <b>Financial liabilities</b>                  |      |                |                |                |                |                |                |
| Borrowings (non-current)                      | 4.6  | 20,528         | 24,318         | 12,961         | 2,897          | 3,009          | 0              |
| Borrowings (current)                          | 4.6  | 6,478          | 1,466          | 13,684         | 7,459          | 1,474          | 515            |
| Derivative financial instruments              | 4.5  | 1,272          | 2,903          | 2,184          | 1,365          | 734            | 2,024          |
| <b>Financial liabilities</b>                  |      | <b>28,278</b>  | <b>28,687</b>  | <b>28,829</b>  | <b>11,721</b>  | <b>5,217</b>   | <b>2,539</b>   |
| <b>Net Financial Obligations (NFO)</b>        |      | <b>-5,549</b>  | <b>1,370</b>   | <b>8,738</b>   | <b>-4,434</b>  | <b>-11,780</b> | <b>-14,545</b> |
| <b>Minority interest (MIN)</b>                |      | <b>0</b>       | <b>0</b>       | <b>0</b>       | <b>0</b>       | <b>0</b>       | <b>0</b>       |
| <b>Common shareholders equity (CSE)</b>       |      | <b>106,561</b> | <b>83,486</b>  | <b>70,746</b>  | <b>66,325</b>  | <b>57,593</b>  | <b>51,839</b>  |

Figure A.2: Reformulated Balance Sheet

Note: All values are in DKK million

Source: [Annual Report 2018]-[Annual Report 2023] and own calculations

|                                                                          |             |               |               |               |               |               |               |
|--------------------------------------------------------------------------|-------------|---------------|---------------|---------------|---------------|---------------|---------------|
| <b>Income statement</b>                                                  |             |               |               |               |               |               |               |
| for the year ended 31 December                                           |             |               |               |               |               |               |               |
| <b>DKK million</b>                                                       | <b>Note</b> | <b>2023</b>   | <b>2022</b>   | <b>2021</b>   | <b>2020</b>   | <b>2019</b>   | <b>2018</b>   |
| <b>Income statement</b>                                                  |             |               |               |               |               |               |               |
| Net sales                                                                | 2.1, 2.2    | 232,261       | 176,954       | 140,800       | 126,946       | 122,021       | 111,831       |
| Cost of goods sold                                                       | 2.2         | -35,765       | -28,448       | -23,658       | -20,932       | -20,088       | -17,617       |
| Gross profit                                                             |             | 196,496       | 148,506       | 117,142       | 106,014       | 101,933       | 94,214        |
| Sales and distribution costs                                             | 2.2         | -56,743       | -46,217       | -37,008       | -32,928       | -31,823       | -29,397       |
| Research and development costs                                           | 2.2, 2.3    | -32,443       | -24,047       | -17,772       | -15,462       | -14,220       | -14,805       |
| Administrative costs                                                     | 2.2         | -4,855        | -4,467        | -4,050        | -3,958        | -4,007        | -3,916        |
| Other operating income and expenses                                      | 2.2, 2.5    | 119           | 1,034         | 332           | 460           | 600           | 1,152         |
| Operating profit                                                         |             | 102,574       | 74,809        | 58,644        | 54,126        | 52,483        | 47,248        |
| Financial income                                                         | 4.10        | 2,945         | 239           | 2,887         | 1,628         | 65            | 2,122         |
| Financial expenses                                                       | 4.10        | -845          | -5,986        | -2,451        | -2,624        | -3,995        | -1,755        |
| Profit before income taxes                                               |             | 104,674       | 69,062        | 59,080        | 53,130        | 48,553        | 47,615        |
| Income taxes                                                             | 2.6         | -20,991       | -13,537       | -11,323       | -10,992       | -9,602        | -8,987        |
| <b>Net profit</b>                                                        |             | <b>83,683</b> | <b>55,525</b> | <b>47,757</b> | <b>42,138</b> | <b>38,951</b> | <b>38,628</b> |
| <b>Earnings per share</b>                                                |             |               |               |               |               |               |               |
| Basic earnings per share (DKK)                                           | 4.1         | 18.67         | 12.26         | 10.40         | 18.05         | 16.41         | 15.96         |
| Diluted earnings per share (DKK)                                         | 4.1         | 18.62         | 12.22         | 10.37         | 18.01         | 16.38         | 15.93         |
| <b>Statement of comprehensive income</b>                                 |             |               |               |               |               |               |               |
| <b>DKK million</b>                                                       | <b>Note</b> | <b>2023</b>   | <b>2022</b>   | <b>2021</b>   | <b>2020</b>   | <b>2019</b>   | <b>2018</b>   |
| <b>Statement of comprehensive income</b>                                 |             |               |               |               |               |               |               |
| Net profit                                                               |             | 83,683        | 55,525        | 47,757        | 42,138        | 38,951        | 38,628        |
| <b>Other comprehensive income</b>                                        |             |               |               |               |               |               |               |
| Remeasurements of retirement benefit obligations                         |             | 13            | 615           | 146           | -67           | -187          | 87            |
| Items that will not be reclassified subsequently to the income statement |             | 13            | 615           | 146           | -67           | -187          | 87            |
| Exchange rate adjustments of investments in subsidiaries                 | 4.3         | -1,404        | 2,289         | 1,624         | -1,689        | 226           | 491           |
| Cash flow hedges:                                                        |             |               |               |               |               |               |               |
| Realisation of previously deferred (gains)/losses                        | 4.3, 4.5    | -1,026        | 1,740         | -1,802        | 329           | 1,677         | -2,027        |
| Deferred gains/(losses) incurred during the period                       | 4.3, 4.5    | 1,612         | 1,026         | -1,755        | 1,384         | -329          | -1,677        |
| Other items                                                              | 4.3         | 4             | -3            | 112           | 10            | 9             | -27           |
| Income tax related to these items                                        | 2.6, 4.3    | -359          | -889          | 1,005         | -577          | -231          | 755           |
| Items that will be reclassified subsequently to the income statement     |             | -1,173        | 4,163         | -816          | -543          | 1,352         | -2,485        |
| <b>Other comprehensive income</b>                                        |             | <b>-1,160</b> | <b>4,778</b>  | <b>-670</b>   | <b>-610</b>   | <b>1,165</b>  | <b>1,165</b>  |
| <b>Total comprehensive income</b>                                        |             | <b>82,523</b> | <b>60,303</b> | <b>47,087</b> | <b>41,528</b> | <b>40,116</b> | <b>36,230</b> |

**Figure A.3:** Reported Income Statement

**Note:** All values are in DKK million

**Source:** [Annual Report 2018]-[Annual Report 2023]

| DKK million                                              | Note     | 2023           | 2022           | 2021           | 2020           | 2019           | 2018          |
|----------------------------------------------------------|----------|----------------|----------------|----------------|----------------|----------------|---------------|
| Net sales                                                | 2.1, 2.2 | 232,261        | 176,954        | 140,800        | 126,946        | 122,021        | 111,831       |
| Cost of goods sold                                       | 2.2      | -35,765        | -28,448        | -23,658        | -20,932        | -20,088        | -17,617       |
| <b>Gross profit</b>                                      |          | <b>196,496</b> | <b>148,506</b> | <b>117,142</b> | <b>106,014</b> | <b>101,933</b> | <b>94,214</b> |
| Sales and distribution costs                             | 2.2      | -56,743        | -46,217        | -37,008        | -32,928        | -31,823        | -29,397       |
| Research and development costs                           | 2.2, 2.3 | -32,443        | -24,047        | -17,772        | -15,462        | -14,220        | -14,805       |
| Administrative costs                                     | 2.2      | -4,855         | -4,467         | -4,050         | -3,958         | -4,007         | -3,916        |
| Other operating income and expenses                      | 2.2, 2.5 | 119            | 1,034          | 332            | 460            | 600            | 1,152         |
| Reversal of D&A                                          |          | 9,413          | 7,362          | 6,025          | 5,753          | 5,661          | 3,925         |
| <b>EBITDA</b>                                            |          | <b>111,987</b> | <b>82,171</b>  | <b>64,669</b>  | <b>59,879</b>  | <b>58,144</b>  | <b>51,173</b> |
| Amortization and depreciation                            |          | -9,413         | -7,362         | -6,025         | -5,753         | -5,661         | -3,925        |
| <b>Core operating profit from sales, before tax</b>      |          | <b>102,574</b> | <b>74,809</b>  | <b>58,644</b>  | <b>54,126</b>  | <b>52,483</b>  | <b>47,248</b> |
| Income taxes                                             | 2.6      | -20,991        | -13,537        | -11,323        | -10,992        | -9,602         | -8,987        |
| Tax benefit                                              |          | 462            | -1,264         | 96             | -219           | -865           | 81            |
| <b>Core operating profit from sales, after tax</b>       |          | <b>82,045</b>  | <b>60,008</b>  | <b>47,417</b>  | <b>42,915</b>  | <b>42,016</b>  | <b>38,342</b> |
| Remeasurements of retirement benefit obligations         |          | 13             | 615            | 146            | -67            | -187           | 87            |
| Tax on other operating profit                            |          | 3              | 135            | 32             | -15            | -41            | 19            |
| <b>Core other operating profit, after tax</b>            |          | <b>10</b>      | <b>480</b>     | <b>114</b>     | <b>-52</b>     | <b>-146</b>    | <b>68</b>     |
| <b>Core operating income, after tax (Core OI)</b>        |          | <b>82,055</b>  | <b>60,487</b>  | <b>47,531</b>  | <b>42,863</b>  | <b>41,871</b>  | <b>38,410</b> |
| <b>Non-core operating income</b>                         |          |                |                |                |                |                |               |
| Exchange rate adjustments of investments in subsidiaries | 4.3      | -1,404         | 2,289          | 1,624          | -1,689         | 226            | 491           |
| Cash flow hedges:                                        |          |                |                |                |                |                |               |
| Realisation of previously deferred (gains)/losses        | 4.3, 4.5 | -1,026         | 1,740          | -1,802         | 329            | 1,677          | -2,027        |
| Deferred gains/(losses) incurred during the period       | 4.3, 4.5 | 1,612          | 1,026          | -1,755         | 1,384          | -329           | -1,677        |
| Other items                                              | 4.3      | 4              | -3             | 112            | 10             | 9              | -27           |
| Income tax related to these items                        | 2.6, 4.3 | -356           | -754           | 1,037          | -592           | -272           | 774           |
| <b>Non-core operating income, after tax</b>              |          | <b>-1,170</b>  | <b>4,298</b>   | <b>-784</b>    | <b>-558</b>    | <b>1,311</b>   | <b>-2,466</b> |
| <b>Net operating income, after tax (OI)</b>              |          | <b>80,885</b>  | <b>64,786</b>  | <b>46,747</b>  | <b>42,305</b>  | <b>43,181</b>  | <b>35,944</b> |
| Financial expenses                                       | 4.10     | -845           | -5,986         | -2,451         | -2,624         | -3,995         | -1,755        |
| Financial income                                         | 4.10     | 2,945          | 239            | 2,887          | 1,628          | 65             | 2,122         |
| <b>Net financial expenses, before tax</b>                |          | <b>2,100</b>   | <b>-5,747</b>  | <b>436</b>     | <b>-996</b>    | <b>-3,930</b>  | <b>367</b>    |
| Tax benefit                                              |          | -462           | 1,264          | -96            | 219            | 865            | -81           |
| <b>Net financial expenses, after tax</b>                 |          | <b>1,638</b>   | <b>-4,483</b>  | <b>340</b>     | <b>-777</b>    | <b>-3,065</b>  | <b>286</b>    |
| <b>Total comprehensive income</b>                        |          | <b>82,523</b>  | <b>60,303</b>  | <b>47,087</b>  | <b>41,528</b>  | <b>40,116</b>  | <b>36,230</b> |

Figure A.4: Reformulated Income Statement

Note: All values are in DKK million

Source: [Annual Report 2018]-[Annual Report 2023] and own calculations

## A.4 Cost of Capital

### A.4.1 Alternative Methodologies to CAPM

Alternative models to consider when estimating the cost of equity are the Fama-French 3 Factor model (FF3F) and the Arbitrage Pricing Theory (APT).

The FF3F builds upon the fundamental concepts of the CAPM, introducing additional risk factors. It is a multi-factor model that extends the CAPM with by a size factor and a value factor. Hereby, the FF3F captures the historical tendency for smaller companies to outperform larger ones, known as the size premium, and the difference in returns between value stocks and growth stocks, defined as the value premium. The model includes two additional beta-values, capturing the sensitivity of the investment to the size and value factor. With the FF3F as with all multi-factor models, the critique of the measurements problems in the CAPM, is also applicable to these models. In accordance with the critique of the CAPM, increasing the number of risk factors and hence also beta's amplifies the compounded estimation error.

An alternative to the CAPM and the FF3F is the Arbitrage Pricing Theory (APT). Unlike CAPM's single-factor focus on market risk, Arbitrage Pricing Theory (APT) embraces a multi-factor approach, assessing the returns as a linear combination of various macroeconomic factors. Research in asset pricing have suggested many relevant factors and found them to explain the returns on some assets. An example of this being McLean & Pontiff (2016) considering 97 of such factors. The inclusion of multiple factors allows for a more thorough assessment of risk, leading to a potentially more accurate estimation of the required return of assets under various economic circumstances. Thus, the strength of the APT lies in its multiplicity, however it does not withstand the challenge of identifying and accurately measuring these factors and their respective betas. In addition, using APT often requires a significant amount of data.

Considering the scope of the paper, the CAPM offers the most appropriate balance between ease of use and explanatory power for the valuation of Novo Nordisk. The subsequent sections will analyze the market risk premium and beta within the CAPM framework, discussing the various ways of estimating the two parameters.

### A.4.2 Market Risk Premium Approaches: Historical MRP & IERP

#### Historical Market Risk Premium

As indicated by the name, the historical market risk premium estimates the return of the market using historical data. It is the most commonly used approach, where the actual returns earned on stocks over a given horizon in the past is estimated in excess to the returns on the risk-free investment. Although this method is fairly simple, premiums estimated in empirical papers may vary substantially. The two main reasons for divergence between estimated premiums specific for this approach are the time horizon used for the estimation and the way returns are averaged over time.

The primary purpose of the CAPM is to estimate the expected future return of an equity. Even if its agreed that historical MRP's provide the best estimates for future MRP, there is no consensus on the time horizon used for the estimation. The rationale for using return data for a period going far back is that it reduces the estimation error of the market risk premium [Koller, 2020]. Research conducted by Dimson, Marsh, and Staunton (2006) advocates for the use of an extended time horizon to estimate the MRP, utilizing 106 years of return data in their estimation. In opposition, others argue that the risk aversion of the average investors changes over time, and that applying a shorter and more recent period presents a more accurate estimate. However, this should be weighed against the greater estimation error due to the shorter period [Damodaran, 2023].

The second reason for divergence in estimations on historical data relates to how the average return is computed. One can either derive the arithmetic average, that is the simple mean of the annual returns, or the geometric average computed as the compounded return. The arithmetic average yields a higher risk premium than the geometric average, but using it to obtain discount rates, that are compounded over time, is inconsistent [Damodaran, 2023]. Table A.1 shows the average excess return of the MSCI World Index on various periods and computational forms. The market risk premium varies from 5.74% to 8.48% depending on the period and computation, hence highlighting the importance and influence of the selection between these.

| Period  | Arithmetic Mean | Geometric Mean |
|---------|-----------------|----------------|
| 10-year | 7.68%           | 6.67%          |
| 20-year | 7.32%           | 5.74%          |
| 50-year | 8.48%           | 6.98%          |

**Table A.1:** Historical Market Risk Premium

**Source:** [MSCI World Index 2024b], [FRED, 2024], [ECB, 2024] and own calculations.

### Implied Equity Risk Premium

In contrast to the historical approach the implied equity risk premium (IERM) utilizes current share prices to estimate the expected market return. The IERM is more forward-looking, and can be estimated using a discounted cash flow-model (DCF). A simple way of doing this is applying the dividend discount model (DDM) as seen below

$$\text{Value of Equity} = \frac{\text{Expected Dividends Next Period}}{(\text{Required Return on Equity} - \text{Expected Growth Rate})} \quad (\text{A.1})$$

This evaluates the present value of dividends growing at a constant rate, thus being a classic stable growth (Gordon) model. Not only does the model assume a constant growth rate but it solely incorporates dividends as the only cash flows to equity. In the last decade dividends made up only half of the free cash flow to equity (FCFE), indicating that the simple DDM-model provides an inaccurate estimate [Damodaran, 2023]. Aswath Damodaran argues for an extension of the model adding share buybacks to the aggregate dividends paid offering a more precise measure of the FCFE. Further, the extended model allows for varying growth rates. Based on these changes, the model can be written as

$$\text{Value of Equity} = \sum_{t=1}^N \frac{E[FCFE_t]}{(1 + k_e)^t} + \frac{E[FCFE_{N+1}]}{(k_e - g_N)(1 + k_e)^N} \quad (\text{A.2})$$

In this equation, the expected free cash flow to equity in year  $t$ , given by  $E[FCFE_t]$ , can grow at a varying rate in the first  $N$  periods, after which it is assumed to grow at a constant rate  $g_N$ . The crucial and unknown parameter  $k_e$  represents the required return on equity.

To employ the two-staged valuation model, the growth rate applied on the FCFE must be determined. Analyst estimates on future growth in earnings and return on equity strongly depend on the time horizon. Forecasts for developed market equities shows an annual return on equity of 7.25% for US dollar investors and 4% for euro investors for the 5 year period from 2023-2027 [Robeco, 2023]. Damodaran argues for the use of growth in earnings for companies in the selected index. The 3-year consensus EPS forecast is a EPS growth of 7.75% for the MSCI World index and 10.05% on S&P 500 [Bloomberg L.P.]. Growth rates at this level cannot be sustained for eternity, hence in line with Damodaran (2023) an initial period of 5 years is applied with higher growth, after which the growth rate is lowered to the risk-free rate of 3.01%. In comparison, the 30-year annualized return forecast on US equity is 3.1% [Schroder PLC]. Since the

long-term forecast aligns closely with the risk-free rate, it supports using the risk-free rate as the constant growth rate in the second stage of the model.

The MSCI World Index was trading at USD 3,169.18 on December 29<sup>th</sup>, 2023. The dividend yield on the index was 1.97% equivalent to a dividend of USD 62.5 per share. To achieve a more accurate estimate, share buybacks should also be considered. However, data on share buybacks for the MSCI World Index is not available, reducing the reliability of the required return derived from this index. Thus, the required return will also be estimated based on the S&P 500 Index. On December 29<sup>th</sup>, 2023, the S&P 500 Index was trading at USD 4,769.83 with combined dividend and share buybacks of USD 160.5 per share [Bloomberg L.P.]. US stocks make up 71% of the MSCI World Index, hence making the S&P 500 index a reasonable comparison [MSCI World Index 2024a]. Using the two-stage growth rates described earlier yields the FCFE for each index, as shown in Table A.2. The initial 5 years are forecasted using the growth estimates of the ones describes in the previous section, where a 10% growth rate is applied.

| Period           | 1      | 2      | 3      | 4      | 5      | Terminal Period | Required return |
|------------------|--------|--------|--------|--------|--------|-----------------|-----------------|
| MSCI World Index | 68.73  | 75.60  | 83.16  | 91.48  | 100.63 | 103.65          | <b>5.79%</b>    |
| S&P 500          | 176.53 | 194.19 | 213.60 | 234.96 | 258.46 | 258.46          | <b>7.75%</b>    |

**Table A.2:** Annual cash flow to equity and estimated required return on indices

**Note:** All cash flows in USD. Table does not show PV of cash flows.

**Source:** [Bloomberg L.P.] & own calculations

The required return is estimated by inserting the cash flows from Table A.2 in Equation (A.2) for each index, applying the risk-free rate of 3.01% as the constant growth rate ( $g_N$ ), and solving for  $k_e$ . The estimated required return on equity is 5.79% and 7.75% for the MSCI World Index and the S&P 500 respectively. As the estimate based on the MSCI World Index omits share buybacks it does not reflect the entire cash flow to equity, hence decreasing the required return on equity. Thus, although the S&P 500 only reflects US stocks, the required return on equity of 7.75% is presumed to be more accurate. As US stocks make up 71% of the MSCI World Index, it is reasonable to apply the S&P 500-estimate to utilize a more valid estimate. Subtracting the risk-free rate from the required return on equity yields a market risk premium of 4.74%.